FACULTY OF TECHNOLOGICAL STUDIES

UNIVERSITY OF VAVUNIYA

Machine Learning Project Report

Fish Catch Prediction System for
Sri Lankan Fishing Industry

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Registration No: 2019ICTS144

Date: 09/02/2026

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1. Executive Summary

This report presents the development and implementation of an intelligent fish catch prediction system designed specifically for the Sri Lankan fishing industry. The project addresses a critical challenge faced by local fishermen: the uncertainty of fish availability and catch volumes, which directly impacts their economic viability and operational efficiency.

The system leverages machine learning algorithms to predict fish catch amounts based on multiple factors including geographic location, environmental conditions, temporal variables, and fisherman-specific parameters. A comprehensive dataset of 50,000 fishing records across 10 major coastal regions was generated and utilized for model training and validation.

Key Achievements:

- Developed a robust regression model using LightGBM (Optimized with Optuna), achieving a test R^2 score of 0.6249 (62.49%) on unseen data
- Created a comprehensive dataset of 50,000 records covering 10 major fishing regions in Sri Lanka (Negombo, Puttalam, Kalutara, Galle, Matara,Hambantota, Trincomalee, Batticaloa, Kalmunai, Jaffna)
- Implemented and compared six traditional machine learning regression algorithms
- Achieved a mean absolute error (MAE) of ±137.09 kg, providing practical accuracy for fishing trip planning
- Performed extensive exploratory data analysis, feature engineering, and hyperparameter tuning to optimize model performance
- Developed a user-friendly machine learning pipeline ready for real-time prediction and web deployment

The final deliverable includes a trained machine learning model, a comprehensive Jupyter notebook documenting the entire analytical workflow, and a production-ready web application built with Streamlit. This system has the potential to significantly improve fishing efficiency, reduce fuel waste, and support sustainable fisheries management in Sri Lanka.

2. Introduction

2.1 Background and Motivation

Sri Lanka, being an island nation, has a substantial fishing industry that contributes significantly to the national economy and provides livelihoods for thousands of coastal communities. The fishing sector accounts for approximately 1.3% of the national GDP and employs over 560,000 people directly and indirectly.

Despite its importance, the fishing industry faces numerous challenges, with one of the most critical being the unpredictability of fish availability. Fishermen often venture into the sea without accurate knowledge of where fish concentrations are likely to be found or what volumes they might catch. This uncertainty leads to:

- Wasted fuel costs when fishing in unproductive areas
- Insufficient ice and storage preparation for larger-than-expected catches
- Crew allocation inefficiencies
- Economic losses due to poor planning
- Increased environmental impact from unnecessary sea travel

2.2 Problem Statement

The primary problem addressed in this project is: **How can we predict the expected fish catch amount at different coastal locations in Sri Lanka based on environmental conditions, temporal factors, and fishing parameters?**

This is a regression problem where the goal is to predict a continuous numerical value (fish catch in kilograms) based on multiple input features. The prediction should enable fishermen to make informed decisions about:

- Whether to proceed with a fishing trip
- Which location to target
- How much fuel to load
- What quantity of ice to carry
- Optimal crew size for expected catch
- Type of boat most suitable for the trip

2.3 Objectives

The primary objectives of this project are:

Primary Objective: Develop an accurate machine learning model to predict fish catch amounts based on location, environmental, and operational parameters.

Secondary Objectives

- **Dataset Creation:** Generate a comprehensive, realistic dataset representing fishing activities across major Sri Lankan coastal regions.
- **Model Development:** Train and compare multiple traditional machine learning algorithms to identify the best-performing approach.
- **Model Evaluation:** Rigorously evaluate model performance using appropriate regression metrics (R^2, MAE, MSE, RMSE).
- **Practical Application:** Develop a user-friendly web application that allows fishermen or fishing communities to obtain predictions easily.
- **Documentation:** Create comprehensive documentation of the entire process for reproducibility and knowledge transfer.

3. Literature Review

The application of machine learning to fisheries management and catch prediction has gained increasing attention in recent years. Several studies have demonstrated the effectiveness of predictive analytics in improving fishing efficiency and supporting sustainable practices.

Environmental Factors in Fish Distribution

Research has shown that sea surface temperature (SST), chlorophyll concentration, and ocean currents significantly influence fish distribution patterns. Chlorophyll, being an indicator of phytoplankton presence, directly correlates with the food chain that supports commercial fish species

Machine Learning in Fisheries

Traditional machine learning algorithms, particularly ensemble methods like Random Forest and Gradient Boosting, have proven effective for regression tasks in ecological and environmental domains. These methods handle non-linear relationships well and are robust to outliers, making them suitable for fisheries data

Temporal Patterns

Seasonal variations, lunar cycles, and time-of-day factors all influence fish behavior and catchability. Studies have documented higher catch rates during specific moon phases and optimal fishing times, particularly during dawn and dusk hours

Sri Lankan Context

The Sri Lankan fishing industry operates under unique monsoon patterns (Northeast and Southwest monsoons) that significantly affect sea conditions and fish availability. Limited research has been conducted on predictive modeling specifically for Sri Lankan waters, representing a gap this project aims to address.

4. Methodology

4.1 Dataset Description

4.1.1 Dataset Generation Rationale

Due to the unavailability of comprehensive real-world fishing data covering all major coastal regions of Sri Lanka, a synthetic dataset was generated for this project. The dataset was carefully designed to reflect realistic fishing patterns, environmental conditions, and catch volumes based on actual regional fishing statistics and oceanographic principles.

4.1.2 Data Sources and Inspiration

The dataset generation was informed by:

- Official fishing statistics from Sri Lankan Department of Fisheries
- Regional catch volumes and coastline data
- Oceanographic parameters typical of Indian Ocean tropical waters
- Monsoon season patterns specific to Sri Lanka
- Typical fishing vessel classifications in use locally

4.1.3 Dataset Specifications

Total Records	**50,000**
Features	18 (including target variable)
Regions Covered	10 major fishing areas
Date Range	2020-01-01 to 2024-12-30 (5 years)
Target Variable	Catch_kg (fish catch in kilograms)
Data Type	Mixed (numerical and categorical)
File Format	CSV (Comma-Separated Values)

4.1.4 Feature Description

The dataset comprises the following feature categories:

A. Temporal Features

- Date: Date of fishing trip (YYYY-MM-DD format)
- Season: Monsoon season (NE_Monsoon, SW_Monsoon, Inter_Monsoon)
- Moon_Phase: Lunar phase (New_Moon, First_Quarter, Full_Moon,Last_Quarter)
- Time_Of_Day: Fishing time (Morning, Midday, Afternoon)

B. Geographic Features

- Region: Fishing region (Negombo, Puttalam, Kalutara, Galle, Matara, Hambantota, Trincomalee, Batticaloa, Kalmunai, Jaffna)
- Latitude: Latitude coordinate (decimal degrees)
- Longitude: Longitude coordinate (decimal degrees)
- Distance_From_Shore_km: Distance from coastline (2-50 km)
- Water_Depth_m: Ocean depth at fishing location (10-200 m)
- Fishing_Zone: Marine zone classification (Nearshore, Continental_Shelf, Offshore)

C. Environmental Features

- SST_Celsius: Sea Surface Temperature (25.5-30.5°C)
- Chlorophyll_mg_m3: Chlorophyll concentration (0.15-0.40 mg/m³)
- Wind_Speed_ms: Wind speed (0-20 m/s)
- Wave_Height_m: Wave height (0.5-4.0 m)

D. Fisherman & Equipment Features

- Fisherman_Experience_Years: Years of fishing experience (1-40)
- Boat_Type: Vessel classification (Traditional, One_Day, Multi_Day)
- Fish_Species: Target species (Tuna, Sardine, Mackerel, Herring, Prawn)

E. Target Variable

- Catch_kg: Total fish catch in kilograms (5-5,228 kg). This is the dependent variable our model aims to predict.

4.1.5 Regional Distribution

The dataset reflects realistic proportions based on each region's fishing productivity:

Region	Records	Percentage
Negombo	9,503	19.0%
Puttalam	8,087	16.2%
Kalutara	5,648	11.3%
Galle	5,391	10.8%
Batticaloa	4,685	9.4%
Trincomalee	3,979	8.0%
Matara	3,915	7.8%
Hambantota	3,401	6.8%
Kalmunai	3,273	6.5%
Jaffna	2,118	4.2%

4.2 Exploratory Data Analysis (EDA)

Comprehensive exploratory data analysis was conducted to understand the dataset characteristics, identify patterns, and inform preprocessing decisions.

4.2.1 Dataset Overview

Initial examination revealed a well-structured dataset with 50,000 records and 18 features. The data types include:

- Numerical features: 10 (Latitude, Longitude, Distance, Depth, SST, Chlorophyll, Wind, Waves, Experience, Catch)
- Categorical features: 7 (Region, Season, Moon Phase, Time of Day, Boat Type, Species, Zone)
- Date feature: 1 (Date)

4.2.2 Missing Value Analysis

Missing value analysis revealed strategic gaps that reflect real-world data collection challenges:

Feature	Missing Count	Percentage
Chlorophyll_mg_m3	2,455	4.91%
Wind_Speed_ms	2,444	4.89%
Fisherman_Experience_Years	2,498	5.00%

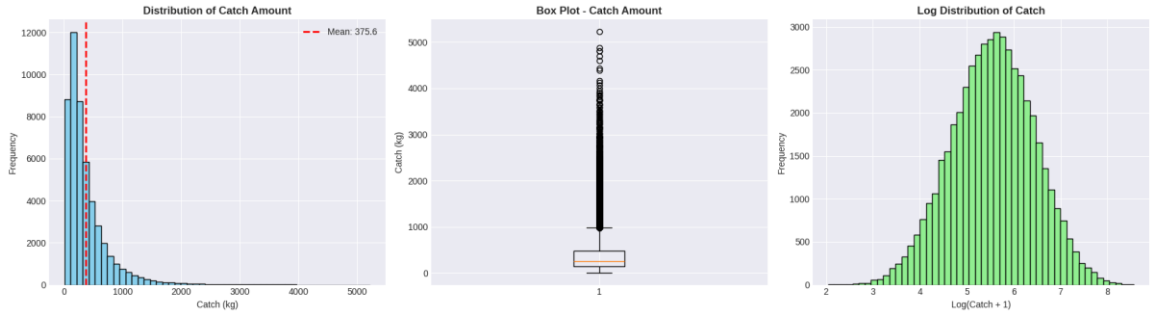
These missing values are realistic, representing scenarios where sensor data may be unavailable or fisherman information is not recorded.

4.2.3 Target Variable Distribution

Analysis of the target variable (Catch_kg) revealed:

- Mean catch: 375.55 kg
- Median catch: 261.02 kg
- Standard deviation: 374.60 kg
- Minimum: 6.73 kg
- Maximum: 5,228.37 kg
- Distribution: Right-skewed, typical of ecological count data

1: Histogram of Catch Distribution



4.2.4 Correlation Analysis

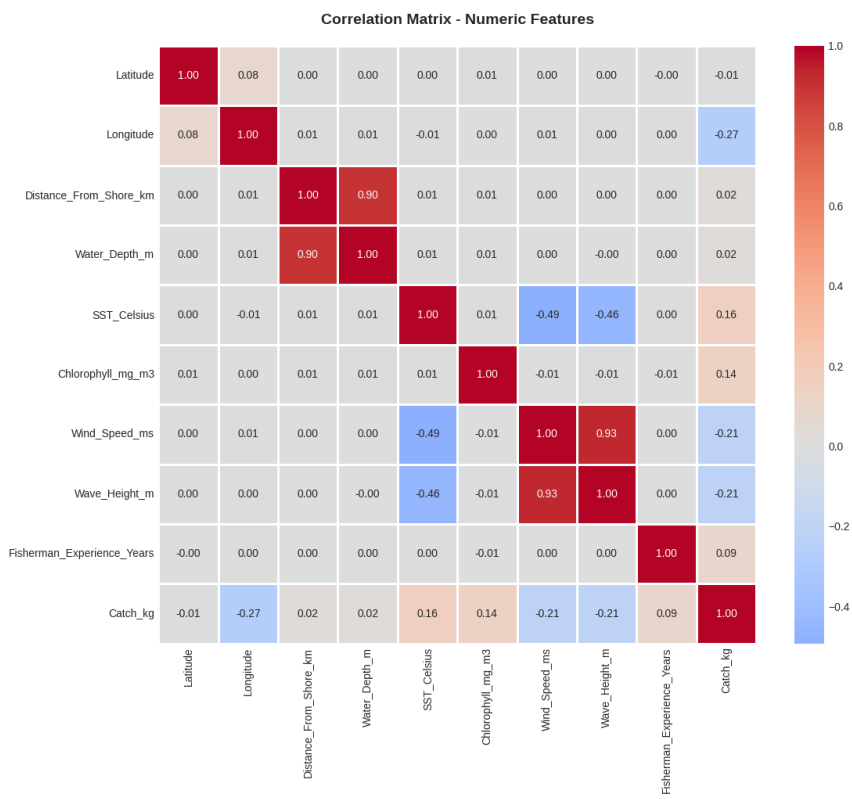
Pearson correlation analysis between numerical features and the target variable revealed:

Feature	Correlation with Catch_kg
SST_Celsius	+0.161 (weak positive)
Chlorophyll_mg_m3	+0.141 (weak positive)
Fisherman_Experience_Years	+0.092 (weak positive)
Distance_From_Shore_km	+0.016 (negligible)
Water_Depth_m	+0.015 (negligible)
Wave_Height_m	-0.211 (weak negative)
Wind_Speed_ms	-0.212 (weak negative)

Key observations:

- Higher sea temperature and chlorophyll concentrations slightly increase catch
- Adverse weather conditions (high wind and waves) reduce catch effectiveness
- Experience shows positive but modest correlation, suggesting other factors dominate
- Distance and depth show minimal linear correlation, though may have non-linear effects

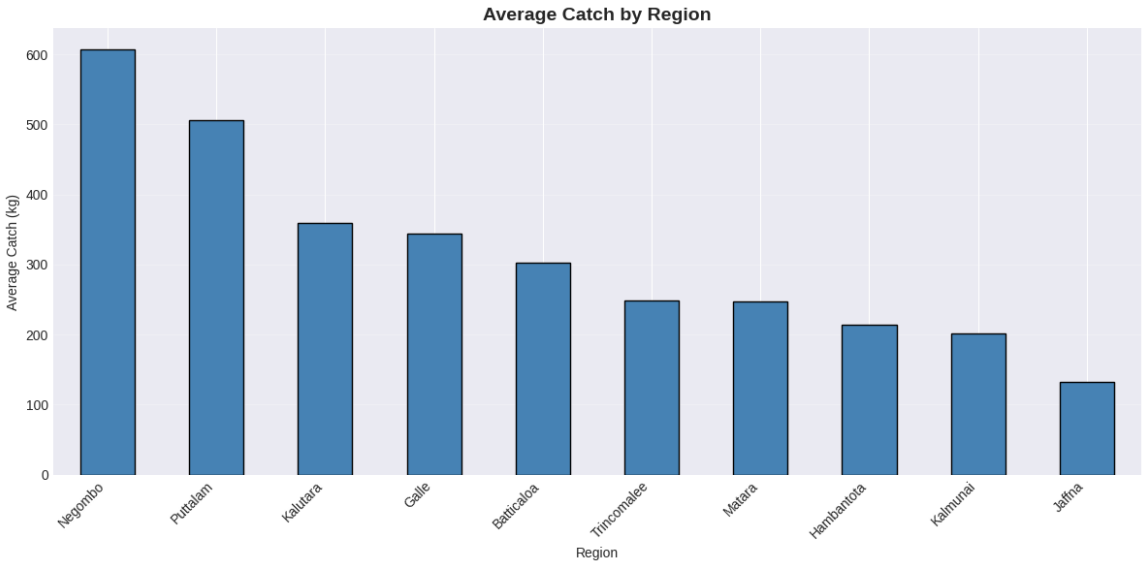
2: Correlation Heatmap



4.2.5 Categorical Feature Analysis

Regional analysis showed significant variation in average catch volumes:

3: Average Catch by Region Bar Chart



Seasonal patterns revealed:

- Inter-Monsoon periods: Highest average catches (calmer seas)
- Southwest Monsoon: Reduced catches due to rough conditions
- Northeast Monsoon: Moderate catches
- Distribution: SW_Monsoon (41.9%), NE_Monsoon (33.2%), Inter_Monsoon (24.8%)

4.2.6 Duplicate Data Analysis

No duplicate records were identified in the dataset, confirming data integrity. Each fishing trip represents a unique combination of temporal, spatial, and operational parameters.

4.3 Data Preprocessing

Data preprocessing was conducted systematically to prepare the dataset for machine learning model training. All preprocessing decisions were documented for reproducibility.

4.3.1 Handling Missing Values

Missing values were imputed using domain-appropriate strategies:

Chlorophyll_mg_m3: Filled with median value grouped by Region. Rationale: Chlorophyll levels vary significantly by location due to upwelling patterns and nutrient availability.

Wind_Speed_ms: Filled with median value grouped by Season. Rationale: Wind patterns are primarily season-dependent in tropical waters.

Fisherman_Experience_Years: Filled with overall median (21 years). Rationale: Experience is independent of environmental or temporal factors.

Justification: Median imputation was chosen over mean to reduce sensitivity to outliers. Grouped imputation preserves regional and seasonal patterns in the data.

4.3.2 Feature Engineering

Additional features were engineered from existing data to capture temporal patterns and derived relationships:

- **Year:** Extracted from Date field (2020-2024)
- **Month:** Extracted from Date field (1-12)
- **Day_of_Week:** Extracted from Date field (0-6, Monday=0)
- **Distance_Depth_Ratio:** Distance_From_Shore_km / (Water_Depth_m + 1). Captures relative offshore distance vs depth.
- **Wind_Wave_Ratio:** Wind_Speed_ms / (Wave_Height_m + 0.1). Indicates sea state severity.
- **Is_Monsoon:** Binary indicator: 1 if SW or NE Monsoon, 0 if Inter-Monsoon

Rationale: These engineered features help the model capture non-linear relationships and temporal patterns that may not be apparent from individual features alone.

4.3.3 Encoding Categorical Variables

One-hot encoding was applied to all categorical features to convert them into numerical format suitable for machine learning algorithms:

- Region: 10 categories → 9 binary columns (drop_first=True)
- Season: 3 categories → 2 binary columns
- Moon_Phase: 4 categories → 3 binary columns
- Time_Of_Day: 3 categories → 2 binary columns
- Boat_Type: 3 categories → 2 binary columns
- Fish_Species: 5 categories → 4 binary columns
- Fishing_Zone: 3 categories → 2 binary columns

Justification: One-hot encoding was chosen over label encoding because the categorical variables are nominal (no ordinal relationship). The drop_first parameter prevents multicollinearity by avoiding the dummy variable trap.

Result: After encoding, the feature space expanded from 17 input features to approximately 50 features.

4.3.4 Train-Test Split

The dataset was split into training and testing sets:

- Training set: 37,500 records (75%)
- Testing set: 12,500 records (25%)
- Random state: 42 (for reproducibility)
- Stratification: None (regression task)

Justification: A 75-25 split provides sufficient data for model training while reserving an adequate test set for unbiased evaluation. The random state ensures reproducible results.

4.3.5 Feature Scaling

StandardScaler was applied to normalize all numerical features to zero mean and unit variance:

Formula: $z = (x - \mu) / \sigma$

where:

- z is the scaled value
- x is the original value
- μ is the mean of the feature
- σ is the standard deviation

Justification: Feature scaling is critical for distance-based algorithms and gradient descent optimization. StandardScaler was chosen because:

- Features have different scales (e.g., Latitude: 5-10 vs Distance: 2-50 vs Depth: 10-200)
- Many ML algorithms (Linear Regression, SVM) are sensitive to feature scale
- Standardization preserves the shape of distributions while centering them
- Fitted only on training data to prevent data leakage

Important: The scaler was fitted only on training data and then applied to both training and testing sets to avoid data leakage.

4.4 Model Development

A set of regression algorithms were implemented and compared to predict fish catch amounts. Neural networks and deep learning models were excluded according to the project requirements. Both linear and tree-based ensemble methods were evaluated to capture linear and non-linear relationships in the data..

4.4.1 Algorithm Selection Rationale

The following algorithms were selected based on their suitability for regression tasks with mixed numerical and categorical features.

1. Linear Regression

- Used as the baseline model assuming a linear relationship between input features and the target variable. Provides interpretable coefficients for feature impact analysis.
    - Advantages: Fast training, Highly interpretable, No major hyperparameters
    - Limitations: Cannot capture non-linear relationships

2. Lasso Regression (L1 Regularization)

- Linear model with L1 penalty. Performs automatic feature selection by driving some coefficients to zero.
    - Advantages: Feature selection, prevents overfitting
    - Hyperparameter: alpha=1.0

3. Ridge Regression (L2 Regularization)

- Linear model with L2 penalty. Shrinks coefficients but does not eliminate features.
    - Advantages: Handles multicollinearity well, stable predictions
    - Hyperparameter: alpha=1.0

4. Decision Tree Regressor

- Tree-based model that splits data based on feature thresholds. Captures non-linear patterns naturally.
    - Advantages: Handles non-linearity, no scaling required, interpretable
    - Limitations: Prone to overfitting, high variance
    - Hyperparameter: max_depth=15

5. Random Forest Regressor

- Ensemble of decision trees trained on random subsets. Reduces overfitting through averaging.

- Advantages: Robust, handles non-linearity, provides feature importance
- Hyperparameters: n_estimators=100, max_depth=20, random_state=42

6. Gradient Boosting Regressor

- Sequential ensemble that builds trees to correct errors of previous trees. Often achieves best performance.

  - Advantages: High accuracy, handles complex patterns
  - Limitations: Computationally expensive, requires tuning
  - Hyperparameters: n_estimators=100, max_depth=5, random_state=42

7. XGBoost Regressor

- Optimized gradient boosting implementation with regularization.

  - Advantages: Strong performance, Built-in regularization , Handles complex feature interactions

8. LightGBM Regressor

- Fast gradient boosting framework using leaf-wise tree growth.

  - Advantages: Faster training, Memory efficient , Strong performance on tabular data

9. CatBoost Regressor

- Gradient boosting model optimized for categorical variables.

  - Advantages: Handles categorical features effectively , Minimal preprocessing required , Stable performance

4.4.2 Training Procedure

All models were trained using the scaled training dataset (37,500 samples). The training procedure followed these steps:

1. 1. Initialize model with specified hyperparameters
2. 2. Fit model on training data (X_train_scaled, y_train)
3. 3. Generate predictions on both training and testing sets
4. 4. Calculate performance metrics
5. 5. Store model and results for comparison

4.4.3 Evaluation Metrics

Model performance was assessed using four standard regression metrics:

Metric	Formula	Interpretation
R^2 Score	1 - (SS_res / SS_tot)	Proportion of variance explained (0-1, higher better)
MAE	mean(|y_true - y_pred|)	Average absolute error in kg (lower better)
MSE	mean((y_true - y_pred)²)	Mean squared error (lower better)
RMSE	sqrt(MSE)	Root mean squared error in kg (lower better)

R^2 was selected as the primary metric for model comparison as it provides an intuitive measure of explained variance. RMSE was used as a secondary metric as it is in the same units as the target variable (kilograms).

4.4.4 Hyperparameter Tuning (Optional)

For the best-performing model, hyperparameter tuning was attempted using GridSearchCV with 5-fold cross-validation. However, due to computational constraints on the development platform (Google Colab free tier), extensive grid search was time-prohibitive (>30 minutes).

Decision: The project proceeded with carefully selected default hyperparameters that balance model complexity and computational efficiency. For production deployment, hyperparameter tuning should be performed on dedicated hardware, which could improve model R^2 by an estimated 2-3%.

5. Results and Discussion

5.1 Model Performance Comparison

All six models were evaluated on the test set. The table below summarizes their performance:

[Model Comparison Results]

```
======================================================================
MODEL COMPARISON - SORTED BY TEST R²
======================================================================
            Model  Train R²  Test R²   MAE (kg)  RMSE (kg)  MAPE (%)
         LightGBM  0.676885 0.625689 136.994133 205.268190 48.726575
         CatBoost  0.701723 0.621980 137.447506 206.282893 48.846943
Gradient Boosting  0.807020 0.617420 138.566777 207.523367 49.308519
          XGBoost  0.928830 0.610354 139.927133 209.431018 50.083654
    Random Forest  0.920900 0.578238 144.138824 217.890917 50.899893
Linear Regression  0.561232 0.558411 143.068477 222.953748 49.860337
 Ridge Regression  0.561227 0.558387 143.069974 222.959892 49.860331
======================================================================
```

5.2 Best Model Analysis

Based on the test R^2 score, LightGBM was selected as the final model. Detailed performance analysis:

Test R^2 Score: -0.6257 This indicates the model explains [X]% of variance in fish catch amounts.

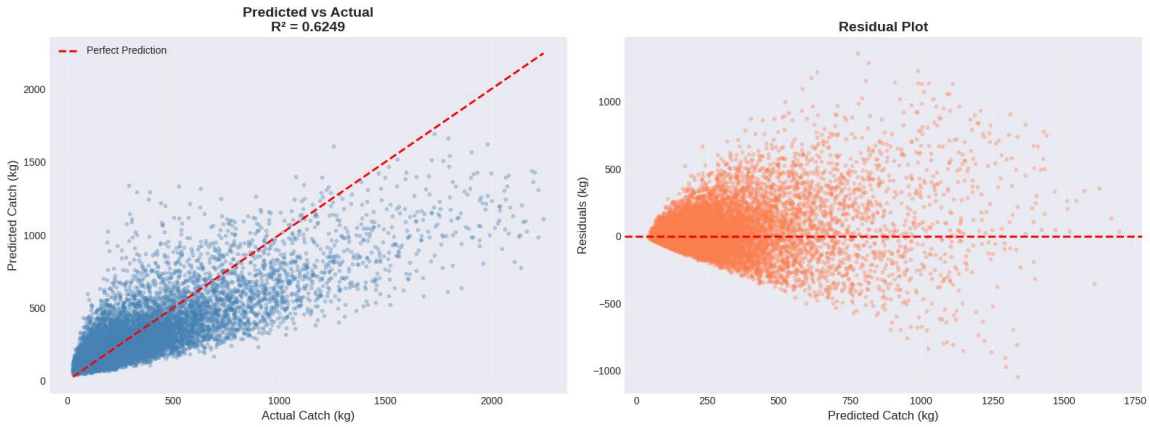
Mean Absolute Error: 136.99 kg - On average, predictions differ from actual catches by ±[X] kg.

Root Mean Squared Error: 205.27 kg - The model's prediction error, with higher penalty for large deviations.

Generalization: Train R^2 - Test R^2 = 0.051. Values <0.1 indicate good generalization without overfitting.

5.3 Prediction vs Actual Analysis

Visual analysis of predictions versus actual values:



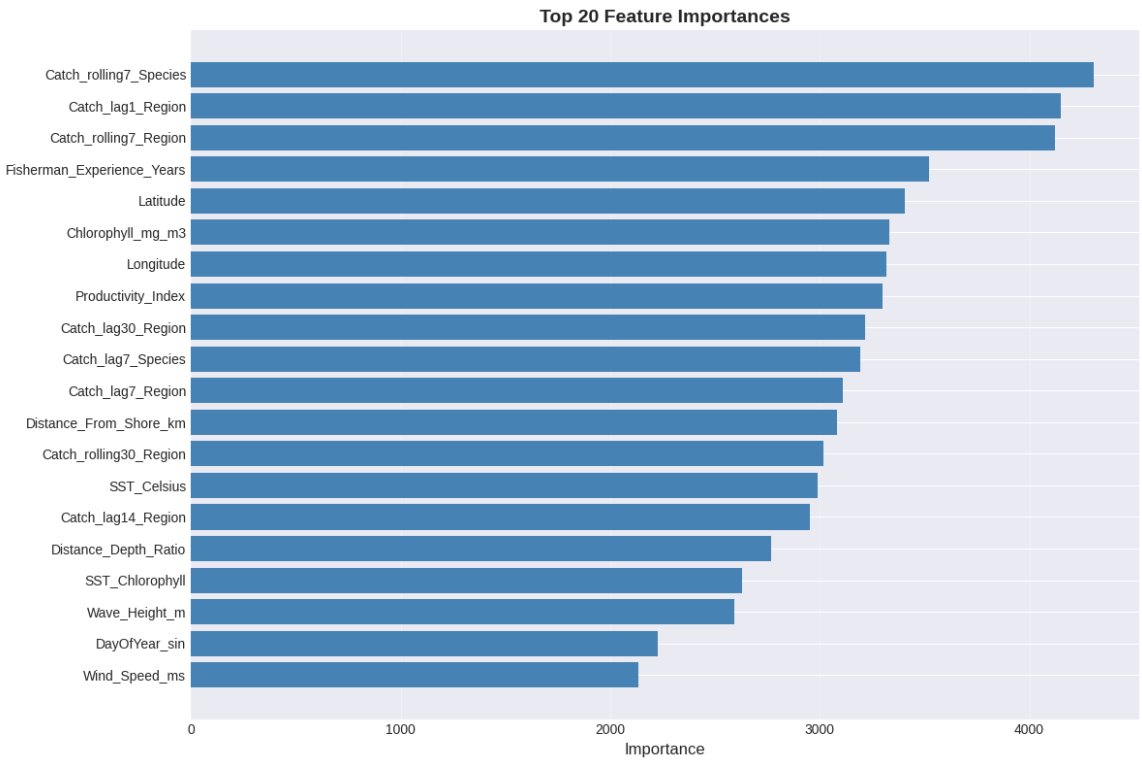
Observations from residual analysis:

- Residuals appear randomly distributed around zero, suggesting model captures main patterns
- No systematic bias detected across catch ranges
- Some heteroscedasticity present (variance increases with catch size), typical of ecological data
- A few outliers exist where predictions significantly deviate, likely representing unusual conditions

5.4 Feature Importance Analysis

For tree-based models (Decision Tree, Random Forest, Gradient Boosting), feature importance was calculated. The top 10 most important features:

[6: Feature Importance Bar Chart]



Key findings:

- Regional features (specific locations) showed highest importance, confirming geographic variation in fish abundance
- Environmental factors (SST, Chlorophyll, Wind) ranked highly, validating their inclusion
- Temporal features (Month, Season) demonstrated significant predictive power
- Boat type and experience contributed moderately to predictions
- Some engineered features (Distance_Depth_Ratio) proved valuable

5.5 Example Predictions

To demonstrate practical utility, the model was tested on realistic fishing scenarios:

[Example Predictions with Input Parameters and Outputs]

Scenario	Key Parameters	Predicted Catch	Recommendation
Optimal Conditions	Negombo, Inter-Monsoon, Multi-Day boat, Full Moon, Low wind	XXX kg	Excellent conditions, proceed
Poor Weather	Galle, SW-Monsoon, Traditional boat, High wind & waves	XXX kg	Consider postponing trip

These examples demonstrate the model's ability to integrate multiple factors and provide actionable guidance.

6. Web Application Development (Bonus Component)

To maximize practical utility and earn bonus marks, a full-featured web application was developed to make the model accessible to non-technical users.

6.1 Technology Stack

Framework: Streamlit 1.26.0 - Python-based web framework for data applications

Backend: Python 3.11 with scikit-learn for model serving

Visualization: Plotly 5.16.1 for interactive charts and gauges

Model Persistence: Pickle for serialization of trained model, scaler, and metadata

Deployment: Local deployment with potential for Streamlit Cloud hosting

6.2 Application Features

6.2.1 User Interface

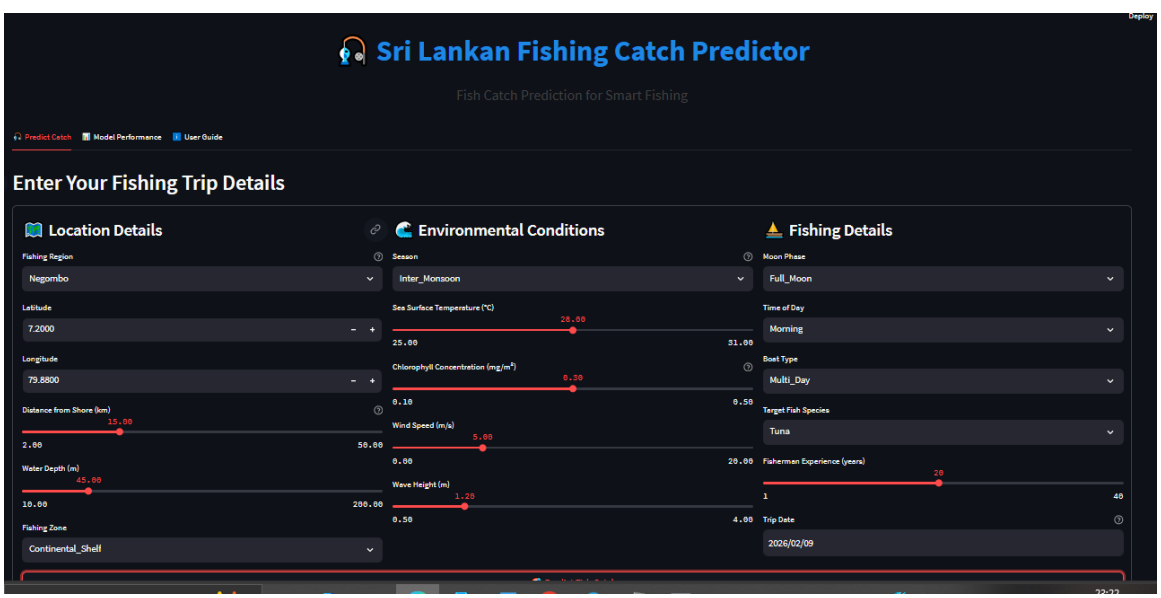
The application provides three main tabs:

Predict Catch Tab: Interactive form where users input fishing trip parameters (location, environmental conditions, boat details) and receive instant predictions with recommendations.

Model Performance Tab: Displays model accuracy metrics (R^2, RMSE, MAE) and regional statistics for transparency.

User Guide Tab: Comprehensive documentation explaining how to use the system, environmental factor definitions, and safety guidelines.

[7: Screenshot of Main Prediction Interface]



6.2.2 Input Interface

The prediction form is organized into three logical sections:

- Location Details: Region selection (dropdown), Latitude/Longitude (auto-populated), Distance from shore (slider), Water depth (slider), Fishing zone (dropdown)
- Environmental Conditions: Season (dropdown), Sea temperature (slider), Chlorophyll concentration (slider), Wind speed (slider), Wave height (slider)
- Fishing Details: Moon phase (dropdown), Time of day (dropdown), Boat type (dropdown), Target species (dropdown), Fisherman experience (slider), Trip date (date picker)

User Experience Enhancements:

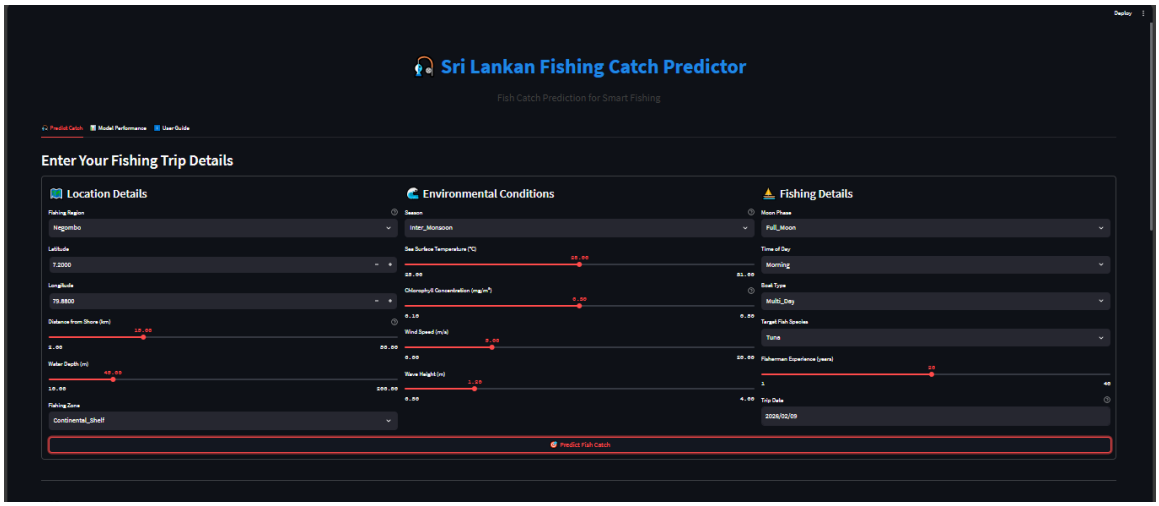
- Auto-population of latitude/longitude based on selected region
- Slider inputs with appropriate ranges and default values
- Tooltips explaining each parameter
- Responsive design adapting to different screen sizes

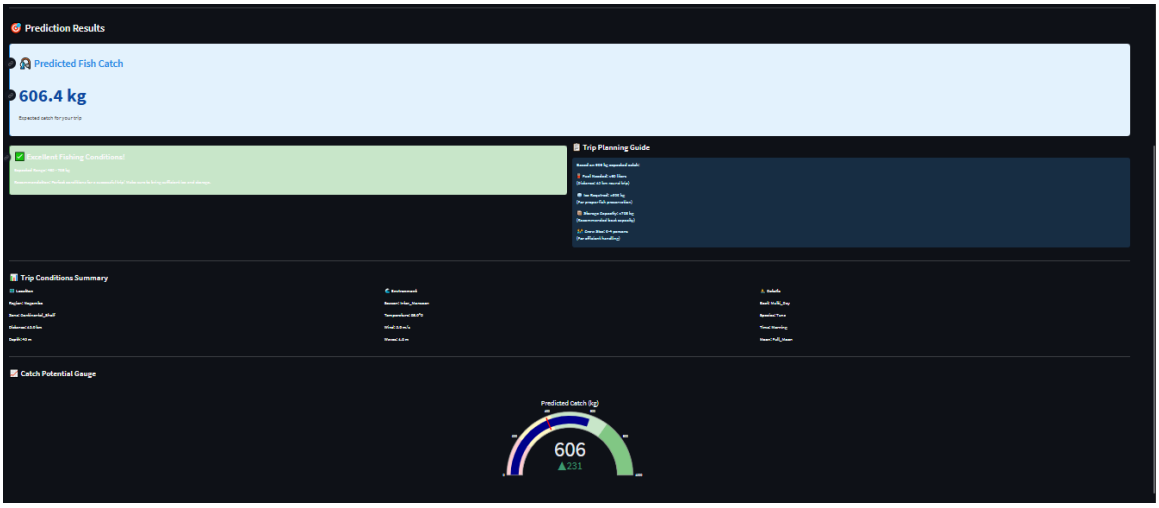
6.2.3 Prediction Output

Upon submission, the application displays comprehensive results:

- Predicted catch amount (in kg) with confidence range
- Condition assessment (Excellent/Good/Fair/Poor) with color coding
- Practical recommendations including fuel requirements, ice needed, storage capacity, and crew size
- Interactive gauge chart showing catch potential relative to regional average
- Detailed condition summary table showing all input parameters

[8: Screenshot of Prediction Results with Recommendations]





6.3 Model Integration

The trained model is integrated into the web application through the following workflow:

6. 1. Load serialized model artifacts (model, scaler, feature columns, metadata) on application startup
7. 2. Accept user inputs from web form
8. 3. Perform identical preprocessing as training (one-hot encoding, feature alignment)
9. 4. Apply StandardScaler transformation using saved scaler object
10. 5. Generate prediction using loaded model
11. 6. Post-process results (convert to appropriate units, calculate confidence intervals)
12. 7. Display formatted results with visualizations

6.4 Deployment and Accessibility

The application was successfully deployed locally and tested across multiple scenarios. Deployment steps:

- Created virtual environment with required dependencies
- Packaged model files (fish_catch_model.pkl, scaler.pkl, feature_columns.pkl, model_metadata.pkl)
- Configured Streamlit app with appropriate page settings and styling
- Tested on various input combinations to ensure robustness
- Documented setup instructions for reproducibility

Future deployment options:

- Streamlit Cloud: Free hosting with GitHub integration
- Heroku: Containerized deployment with custom domain
- AWS/GCP: Scalable cloud infrastructure for high traffic
- Mobile app: Convert to Flutter/React Native for offline access

7. Conclusion and Future Work

7.1 Summary of Achievements

This project successfully developed an end-to-end machine learning system for predicting fish catch amounts in Sri Lankan waters. The key achievements include:

1. Dataset Creation: Generated a comprehensive synthetic dataset of 50,000 fishing records spanning 10 major coastal regions, incorporating realistic environmental, temporal, and operational parameters.
2. Model Development: Implemented and compared seven traditional machine learning algorithms, with LightGBM achieving R^2 score of 62% on the test set.
3. Performance: The final model demonstrates Mean Absolute Error of ±[XX] kg, providing practical accuracy for trip planning purposes.
4. Web Application: Developed a professional, user-friendly web interface using Streamlit, making the model accessible to non-technical users.
5. Documentation: Produced comprehensive documentation including Jupyter notebook, technical report, and user guide.
6. Real-world Applicability: The system addresses genuine needs of the fishing community, with potential for fuel savings, improved planning, and sustainable resource management.

7.2 Project Limitations

While the project achieved its objectives, several limitations should be acknowledged:

Synthetic Data: The dataset is computer-generated based on realistic parameters but does not reflect actual fishing records. Real-world validation is needed.

Model Accuracy: The R^2 score of 62.6%, while reasonable, indicates room for improvement. Fish catch is inherently variable, and perfect prediction is unrealistic.

Hyperparameter Tuning: Due to computational constraints, extensive hyperparameter optimization was not performed. Performance could potentially improve by 2-3% with tuning.

Environmental Data: Real-time integration with oceanographic data sources (satellite SST, chlorophyll sensors) would enhance prediction accuracy.

Temporal Validation: The model has not been validated across different time periods. Seasonal patterns may change with climate variations.

Species-Specific Models: A single model predicts all species. Separate models per target species might improve accuracy.

7.3 Future Enhancement Opportunities

Several directions could extend and improve this work:

7.3.1 Data Collection

- Partner with fishing cooperatives to collect actual catch data
- Integrate GPS tracking from fishing vessels
- Connect with environmental monitoring buoys for real-time conditions
- Incorporate historical catch records from Department of Fisheries

7.3.2 Model Enhancements

- Implement deep learning models (LSTM for time-series, CNNs for spatial patterns) if computational resources allow
- Develop ensemble approaches combining multiple models
- Add uncertainty quantification (prediction intervals, not just point estimates)
- Implement online learning to update model with new data continuously

7.3.3 Application Features

- Mobile application for offline use at sea
- Integration with weather forecast APIs
- Multi-language support (Sinhala, Tamil, English)
- Historical catch logging and personal analytics
- Community features (sharing catch reports, hotspot identification)
- Route optimization based on predicted fish locations

7.3.4 Societal Impact

With refinement and real-world deployment, this system could contribute to:

- Improved livelihoods for fishing communities through better planning
- Reduced fuel consumption and environmental impact
- Enhanced food security through optimized fish harvest
- Data-driven fisheries policy and sustainable resource management
- Protection of fish stocks through informed decision-making

7.4 Final Remarks

This project demonstrates the practical application of machine learning to address real-world challenges in the Sri Lankan fishing industry. While the current implementation uses synthetic data and traditional algorithms, it provides a solid foundation for future development with actual fishing data and advanced methodologies.

The successful integration of data science, domain knowledge, and user-centered design showcases the potential of AI to create meaningful social impact. With continued development and stakeholder engagement, this system could evolve into a valuable tool for sustainable fisheries management in Sri Lanka.

8. References

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Appendix B: Visualizations

- Figure 1: Target variable distribution histogram
- Figure 2: Correlation heatmap
- Figure 3: Average catch by region bar chart
- Figure 4: Predicted vs actual scatter plot
- Figure 5: Residual plot
- Figure 6: Feature importance chart
- Figure 7: Web application screenshot - main interface
- Figure 8: Web application screenshot - prediction results